2026 Spring

Course ID: 1142-1765

AdvancedProgramming

02-Python programming for 'Two Sum'

Advanced Programming

Python · C++ · Rust

Data Structures & Algorithms

Python

C++

Rust

Dynamic Programming · Divide & Conquer · Greedy Algorithms · Arrays · Lists · Maps & Sets

Part-1

Python Basics

LeetCode Problem
"Two Sum"

Course Activities
Github repository
Discussion Forum


🏆 Top 10 Programming Languages
Source: TIOBE Index · November 2025
🥇 Python
23.37%
🥈 C++
10.03%
🥉 Java
9.45%
4 C
8.89%
5 C#
6.73%
6 JavaScript
3.79%
7 Go
2.27%
8 Rust 🦀
1.98% ↑
9 Swift
1.54%
10 TypeScript
1.42%
0%
5%
10%
15%
20%
23%
🦀 Rust reached its all-time high of #8 in Nov 2025 — memory safety & performance driving adoption

Python — Advanced Features

Comprehensions & Generators

List/dict/set comprehensions, generator expressions for memory-efficient iteration

Decorators & Context Managers

Higher-order functions, @property, @staticmethod, with statement patterns

Type Hints & Dataclasses

PEP 484 type annotations, @dataclass, Protocol for structural subtyping

Python

```python
# Generator + type hints + dataclass
from dataclasses import dataclass
from typing import Iterator

@dataclass
class Node:
    val: int
    children: list["Node"]

def traverse(root: Node) -> Iterator[int]:
    yield root.val
    for child in root.children:
        yield from traverse(child)

# Lazy evaluation — memory efficient
tree = Node(1, [Node(2, []), Node(3, [])])
result = list(traverse(tree))  # [1, 2, 3]
```

Key Concepts

- GIL & asyncio concurrency
- Magic methods (__dunder__)
- Metaclasses & ABCs
- functools: lru_cache, partial
- itertools: chain, islice, product
- Walrus operator :=

⌨️ Keyboard Shortcuts Cheat Sheet

📓 Google Colab

`Ctrl+Enter`	Run cell
`Shift+Enter`	Run + next cell
`Alt+Enter`	Run + insert below
`Ctrl+M B`	Insert cell below
`Ctrl+M D`	Delete cell
`Ctrl+M M`	Cell to Markdown
`Ctrl+M Y`	Cell to Code
`Ctrl+F9`	Run all cells

Part-2

Python Basics

LeetCode Problem
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LeetCode Practice

Master Programming Skills with Python · C++ · Rust

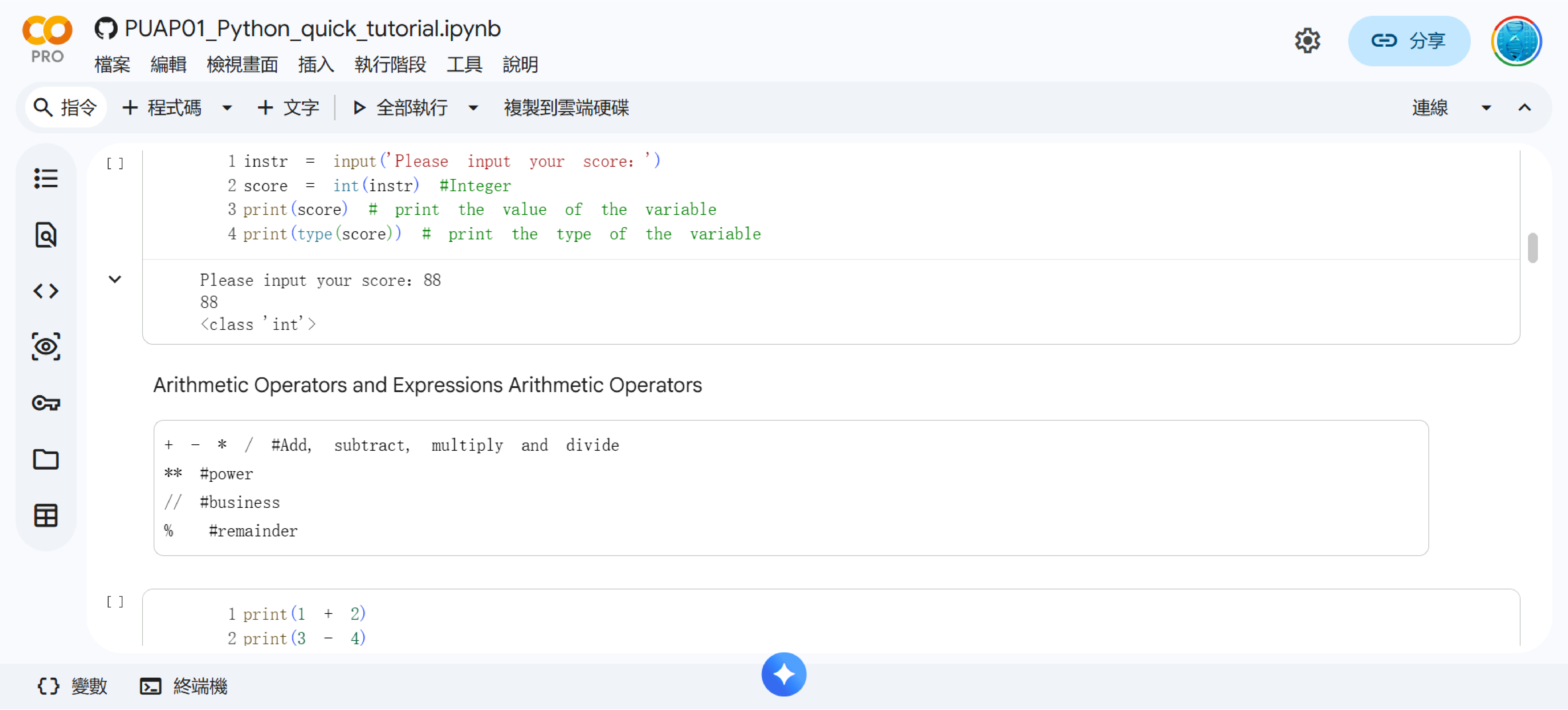
2500+	50+	10M+
Problems	Companies	Users

Difficulty Tiers

- Easy
- Medium
- Hard

Arrays · Strings · Trees · Graphs · DP · Greedy · Divide & Conquer · Two Pointers · Sliding Window

Two Sum



https://leetcode.com/problems/two-sum/description/

https://leetcode.cn/problems/two-sum/description/

Part-3

Python
Basics

LeetCode Problem
"Two Sum"

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Creating a

GitHub Repository

Remote Storage

Host code in the cloud

Collaboration

Work with any team

Version Control

Track every change

Step-by-Step Guide for Beginners · git init → push → collaborate

1 Sign Up & Set Up Your Account

Go to github.com and register

Enter your details
Username, email address, and a strong password. Your username will be part of your public URL.

Verify your email
GitHub sends a confirmation email. Click the link to activate your account before proceeding.

Choose a plan
Free tier is sufficient for most students. Public repos are unlimited; private repos allowed on free plan.

Enable 2FA (recommended)
Two-factor authentication adds security. Use an authenticator app like Google Authenticator.

Your GitHub profile URL will be:

`github.com/your-username`

Username Tips

- Use lowercase letters and hyphens only
- Keep it professional — it appears on your resume
- Avoid numbers at the end (e.g., user123)
- Make it short and memorable

2 Create a New Repository

github.com/new

Create a new repository

Owner

your-username

Repository name *

my-first-repo

Description (optional)

My first GitHub project

Visibility

Public

Private

Repository Name

Must be unique within your account. Use lowercase, hyphens, no spaces. e.g. my-cool-project

Public vs Private

Public: anyone can view. Private: only you and invited collaborators. Students often use Public for portfolios.

Description

Optional but recommended. A brief description helps others understand your project's purpose.

Tip

Click the green '+' button (top-right on GitHub) or go to github.com/new to start creating a repo.

3 Initialize Repository Files

README.md

Highly Recommended

The front page of your repository. Written in Markdown, it describes what the project does, how to install it, how to use it, and how to contribute.

````
# My Project

A brief description.

## Installation
```bash
npm install
```
````

.gitignore

Recommended

Tells Git which files to NOT track (e.g., node_modules/, .env, __pycache__). GitHub offers templates for Python, Node.js, Java, etc.

```
# Python example
__pycache__/
*.pyc
.env
venv/
.DS_Store
```

LICENSE

Optional

Defines how others may use your code. MIT License is the most permissive and widely used. Required if you want others to legally reuse your work.

```
MIT License

Copyright (c) 2025
Your Name

Permission is hereby
granted, free of charge...
```

4 Clone the Repository to Your Computer

Copy the Clone URL from GitHub: `https://github.com/username/my-first-repo.git`

HTTPS vs SSH

HTTPS (Recommended for beginners)

`https://github.com/user/repo.git`

Use your GitHub username & Personal Access Token as password.

SSH (For advanced users)

`git@github.com:user/repo.git`

Requires SSH key setup but no password prompt after configuration.

Terminal

```
$ git clone https://github.com/
    username/my-first-repo.git
  Cloning into 'my-first-repo'...
  remote: Enumerating objects: 3...
  Receiving objects: 100% (3/3)
$ cd my-first-repo
$ ls
  README.md  .gitignore  LICENSE
```